

UNIT IV
ASSIGNMENT

Class #7: Problems on Properties of eigenvalues and Eigenvectors

1) Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. Verify that the trace equals the sum of eigenvalues and the determinant equals their product. If we shift A to $A - 8I$ what are the eigenvalues and eigenvectors and how are they related to those of A ?

ANS: Eigen Values are 3, 2

Eigen vectors are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Eigen vectors are same and eigen values are been shifted λ to $\lambda - 8$

2) Find the eigenvalues and eigenvectors of A, A^2, A^{-1} and $A + 4I$ if

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$$

ANS: Eigen values are 1,5

Eigen Vectors are

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} k_1$$

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} k_2$$

the eigen values of $A^{-1} = 1, \frac{1}{5}$

The eigen values of A^2 is 1,25.

Eigen value of $A + 4I$: 5,9.

3) Write three different 2 x 2 matrices for which the eigenvalues are 1, 3 and determinant is 3.

$$\text{ANS: } \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}$$